MLI

The MetaLogic Inference Program
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Table of Contents

Introduction 1

Conditions for Using MLI 3

GNU GENERAL PUBLIC LICENSE 5

1 The Concepts of MLI 9

2 Examples 13

3 MLI Input Files 19

4 User Interface 21

5 The MLI Inference Algorithm 23

6 Understanding and Debugging MLI Source Code 25

7 Invoking MLI 27

Appendix A MLI Symbols 29

Appendix B Glossary 31

Bibliography 33

Index of Terms 35

Introduction

MLI

Conditions for Using MLI

GNU GENERAL PUBLIC LICENSE

Preamble

TERMS AND CONDITIONS

END OF TERMS AND CONDITIONS

How to Apply These Terms to Your New Programs

```
one line to give the program's name and a brief idea of what it does.
yearname of author
```

```
programyearname of author
```

1 The Concepts of MLI

1.1 Metatheory and Object Theory

object theorymetatheory

naive set theoryaxiomatic set theorytheories

consistent

first order logicquantifiers

1.2 Formal Theories and Formal Proofs

formal theory(well-formed) formulasrules of inferencepostulaterelations

direct consequence

axiom rulehead bodyclause

first-depth

first-breadth

modus ponensMP

formal proof provable

generalization axiom

1.3 Proof Trees and Unification

unification

goals

free inKleenean

1.4 The Propositional Calculus

propositional calculusmodus ponensquantifierstautologymetatheoremtruth table

excluded middleintuitionistic propositional calculusHeyting algebra

Ł*logical negationlogical implicationconsistentlogical andlogical or logical equivalencedefinitionslogical truelogical falseatomsdeduction theoremmetatheorem*

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classical systemintuitionistic system

1.5 The Predicate Calculus (Quantification Theory)

1.6 Theory with Equality

1.7 Formal Number Theory

1.8 Axiomatic Set Theory

2 Examples

2.1 Simple Propositional Calculus

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2.1.1 Nested proofs

depth

3 MLI Input Files

4 User Interface.

5 The MLI Inference Algorithm.

6 Understanding and Debugging MLI Source Code.

7 Invoking MLI

infile
infile
infile

7.1 MLI Options

file	*filefilefilename basefilename base*
file	*filefilename base*
file	*filefilename base*
dir	*dirdir*
name	*namefilename base*

7.2 Option Cross Key

Appendix A MLI Symbols

Symbol]

Appendix B Glossary

Bibliography

Index of Terms